

HealthLedger AI

Accounting Intelligence for Healthcare

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1. FIELD OF THE INVENTION

The present invention relates to artificial intelligence systems applied to healthcare financial administration, and more particularly to an AI-powered compliance and decision-support system specifically designed for hospital accounting departments, trained on institution-specific cooperation agreements and operative in real time at the point of accounting decision.

2. BACKGROUND OF THE INVENTION / PROBLEM STATEMENT

Hospital accounting departments operate within a uniquely complex regulatory and contractual environment. The following problems motivate the present invention:

■ **Volume and Complexity of Cooperation Agreements:** A single hospital network may maintain in excess of 200 active cooperation agreements with insurers, pharmaceutical suppliers, medical device manufacturers, and other healthcare entities. Each agreement contains clauses that carry direct accounting obligations — revenue recognition triggers, cost allocation rules, and compliance thresholds.

■ **Inconsistent Interpretation:** Agreement clauses are routinely interpreted differently across accounting staff, teams, and fiscal periods. There is no authoritative, machine-readable source of truth. This leads to divergent booking decisions that create audit exposure and regulatory risk.

■ **Financial Exposure from Clause Misinterpretation:** A single misread clause in a cooperation agreement can create liability in the range of tens of thousands to hundreds of thousands of euros. Current tools — general-purpose ERP systems, document management platforms, and generic AI assistants — are not trained on, nor capable of reasoning about, the specific contractual language of a given institution.

■ **Absence of Domain-Specific AI:** No commercially available AI system has been built specifically for hospital cooperation agreement compliance. Existing systems lack the healthcare accounting domain knowledge, the institution-specific training data, and the clause-traceable response architecture required for deployment in a clinical finance environment.

■ **Gap Between Legal Text and Accounting Action:** The translation from contractual clause to concrete accounting entry requires expert interpretation that is currently manual, time-consuming, and error-prone.

3. SUMMARY OF THE INVENTION

HealthLedger AI is an artificial intelligence system comprising a managed AI agent architecture trained on hospital-specific cooperation agreements and deployed as a real-time compliance and

decision-support tool for healthcare accounting departments.

The invention is grounded in original doctoral research and built through an on-site immersion methodology — a novel approach in which the system is developed in direct collaboration with hospital finance teams, capturing institution-specific accounting conventions, interpretive practices, and compliance workflows that cannot be derived from public data alone.

The invention is characterized by four core pillars:

Pillar 1: Cooperation Agreement Mapping

Automated parsing, semantic segmentation, and structured mapping of all clauses within hospital cooperation agreements, identifying accounting obligations, revenue thresholds, and compliance triggers.

Pillar 2: Accounting Compliance Intelligence

A real-time question-answering and guidance layer that delivers compliance determinations tied directly and traceably to source clauses, enabling accountants to make defensible decisions at the point of action.

Pillar 3: Hospital-Specific AI Model

A domain-adapted AI model trained on the institution's own chart of accounts, internal naming conventions, team-specific interpretive practices, and historical accounting decisions — creating an AI that speaks the language of the institution.

Pillar 4: On-Site Immersion Methodology

A proprietary methodology for AI system development through embedded collaboration with hospital accounting teams, yielding training data and institutional knowledge that cannot be replicated through remote or generic data acquisition.

4. DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

4.1 System Architecture Overview

In its preferred embodiment, the invention is implemented as a managed AI agent system deployed over a secure API infrastructure. A persistent agent entity is created with a fixed identity and domain-specific instruction set. Each user interaction is conducted through a session abstraction layer, enabling stateful, context-aware exchanges. The underlying language model is a large-scale transformer architecture, such as Claude Opus, accessed via the Anthropic Managed Agents API.

4.2 Step 01 — Agreement Ingestion

Upon onboarding, all cooperation agreements of the target hospital network are ingested into the system. The ingestion pipeline performs: (a) document parsing and clause segmentation; (b) semantic tagging of clause types (revenue recognition, cost allocation, compliance threshold, penalty trigger, etc.); (c) structured indexing into a vector database for retrieval-augmented generation (RAG); and (d) metadata annotation linking each clause to the parent agreement,

effective dates, and responsible accounting unit. The output is a machine-queryable knowledge base of the institution's contractual obligations.

4.3 Step 02 — Domain Training

The AI agent is trained and/or fine-tuned on institution-specific accounting materials, including: the hospital's chart of accounts; internal accounting policies and procedure manuals; historical booking decisions and their justifications; and the interpretive conventions of the finance team as captured during on-site immersion. This training step produces a model that correctly uses the institution's own terminology and is calibrated to its specific compliance posture. Domain training may be implemented via supervised fine-tuning, in-context learning, or instruction-tuned prompting augmented by RAG retrieval.

4.4 Step 03 — Live Compliance Guidance

In production, the system operates as a real-time advisory interface accessible to accountants at the point of decision. When an accounting question is submitted — e.g., 'Under Agreement X, how should we recognize revenue from Service Y in Q3?' — the system: (a) retrieves the most relevant clauses from the agreement knowledge base via semantic search; (b) reasons over those clauses in the context of the query; (c) generates a compliance determination with a specific accounting recommendation; and (d) returns a response that explicitly cites the source clause and agreement, making the determination fully auditable and defensible. Response latency is designed to be below five seconds for standard queries.

4.5 On-Site Immersion Methodology

A distinguishing feature of the invention is its development methodology. Unlike AI systems built on publicly available or synthetic data, HealthLedger AI is constructed through direct, sustained engagement with hospital accounting teams in their working environment. This immersion yields: (a) ground-truth training data reflecting real institutional decisions; (b) identification of tacit knowledge and informal interpretive conventions not captured in written documents; and (c) iterative validation of system outputs against expert judgement, prior to deployment. This methodology constitutes a novel approach to domain-specific AI development in regulated healthcare environments.

5. DRAFT CLAIMS

Note: The following claims are presented in preliminary form for discussion with patent counsel. Final claim language and scope will be determined in coordination with the prosecuting attorney.

Claim 1 — Independent (System)

A computer-implemented artificial intelligence system for healthcare accounting compliance, comprising: a document ingestion module configured to parse and semantically segment cooperation agreements of a healthcare institution; a structured knowledge base storing agreement clauses indexed by accounting obligation type; a domain-adapted language model trained on institution-specific accounting terminology, chart of accounts, and interpretive conventions; a retrieval-augmented generation layer configured to retrieve relevant clauses in response to accounting queries; and a compliance output module configured to generate accounting guidance with traceable citation of source clauses.

Claim 2 — Independent (Method)

A method for generating real-time accounting compliance guidance in a healthcare institution, comprising: ingesting cooperation agreements and extracting accounting-relevant clauses therefrom; training an artificial intelligence model on institution-specific accounting data acquired through on-site immersion with the institution's finance team; receiving an accounting query from a user; retrieving one or more relevant agreement clauses using semantic similarity search; and generating a compliance determination responsive to the query, wherein the determination includes explicit citation of the retrieved clause and the source agreement.

Claim 3 — Dependent on Claim 1

The system of Claim 1, wherein the domain-adapted language model is further trained on historical accounting decisions and justifications recorded during on-site collaboration with the institution's accounting personnel.

Claim 4 — Dependent on Claim 1

The system of Claim 1, wherein compliance output includes a confidence score and a reference to the specific clause number and section of the source cooperation agreement.

Claim 5 — Dependent on Claim 2

The method of Claim 2, wherein on-site immersion comprises sustained embedded engagement with the institution's accounting department to capture tacit interpretive conventions not represented in written documentation.

Claim 6 — Dependent on Claim 1

The system of Claim 1, wherein the system is implemented as a managed AI agent with a persistent agent identity, and wherein each user interaction is mediated through a discrete session layer enabling stateful, context-aware exchanges.

Claim 7 — Independent (Use Case)

A use of an artificial intelligence system as described in Claim 1, for the purpose of reducing financial exposure arising from misinterpretation of cooperation agreement clauses in hospital accounting operations.

6. ABSTRACT

HealthLedger AI is an artificial intelligence system providing real-time accounting compliance intelligence for hospital finance departments. The system ingests and semantically indexes a hospital network's cooperation agreements, trains a domain-adapted language model on institution-specific accounting data acquired through on-site immersion, and deploys a retrieval-augmented question-answering interface that delivers clause-traceable compliance guidance at the point of accounting decision. The invention addresses the absence of AI tooling capable of reasoning over hospital-specific contractual obligations, reducing financial exposure from clause misinterpretation and enabling defensible, auditable accounting determinations. The system's development methodology — on-site immersion with hospital accounting teams — constitutes a further inventive contribution to the field of domain-specific AI development.

7. INVENTOR DECLARATION

I, the undersigned, declare that I am the original inventor of the subject matter described in this disclosure, that all information provided herein is true and accurate to the best of my knowledge, and that I have not previously disclosed this invention in any manner that would prejudice the filing of a patent application.

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Date: June 26, 2026

Signature:
